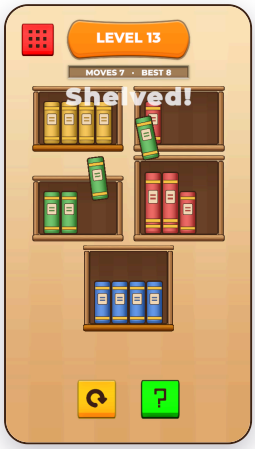
1 · THE AD

A 15-second storyboard

Portrait 9:16, readable with the sound off: the satisfying finish first, then the rule, then height, the cart and the lock one beat each, then a challenge the viewer wants to answer.

0–2 s · Hook



Two green books fly home and a shelf finishes.

SORT THE SHELVES

Sound: Ticks, then a chime

2–4 s · The rule



A full shelf of one colour turns gold: Shelved!

ONE COLOUR PER SHELF

Sound: Chime

4–6.5 s · The twist



A run with a tall atlas bumps against a low shelf and is refused.

TALL BOOKS NEED TALL SHELVES

Sound: Thud

6.5–8.5 s · The cart



A book offered to the returns cart is refused: it only gives.

EMPTY THE CART

Sound: Thud

8.5–10.5 s · The lock



The glass case stays shut; its tag is red.

FINISH RED TO OPEN IT

Sound: Padlock rattle

10.5–12 s · Opened



Red is done, the case is open, and the last blue goes in.

OPENED!

Sound: Click, chime

12–13.5 s · Challenge



Level 40: tall shelves, the cart and a locked case at once.

BEST: 12 MOVES. CAN YOU?

Sound: Music drops

13.5–15 s · End card



Sorted! Three stars; the edit lays the title and a Play button over it.

SHELF LIFE · PLAY FREE

Sound: Win jingle

Every frame is real. A PlayMode test (`AdStoryboard_CapturesTheBeats`) plays levels 4, 13, 21, 31 and 40 with the solver's moves, tries one move each rule refuses, and saves a frame after every move; these are picked from them. The challenge line quotes level 40's real par.

2 · THE CPI TEST

Three hooks, one decision

A hyper-casual prototype's first question is not "is it fun?" but "will people install it cheaply enough?"

Creative variants

VARIANT	FIRST 2 SECONDS	WHAT IT TESTS
A · Shelved (above)	Books fly home and a shelf turns gold	The genre's satisfying sort; the widest audience
B · Too tall	An atlas bumps a low shelf: "Where does it go?"	Is height, the game's difference, the hook?
C · Challenge	A full library and the best number of moves	Puzzle players who want to beat a number

Test design

	SETTING (PLANNING ASSUMPTIONS)
Channels	Meta (Facebook/Instagram) first; an ad network after a pass
Geo and build	US Android, the APK behind a store listing with analytics on
Rounds	Up to three creative rounds before the kill decision; change the hook first, as it is the cheapest
Reading	Compare variants only once each has enough impressions for its CTR to settle

Go / iterate / kill

Go

- CPI under $0.30 [1]
- CTR above 2% [2]
- IPM above 80 on Android [2]
- D1 at least 38% on Android [1]
- Day-7 playtime per user over 2,000 s [1]

Soft launch with the KPI plan overleaf.

Iterate

- CTR good but CPI over the line: the store page loses them (icon, first screenshot)
- CPI good but D1 short: the first minute (levels 1–5) loses them
- Neither: try the next hook

Kill

- After three creative rounds, no variant within reach of the go line on CPI
- Or D1 stays near the genre median (19% on Android [3]) even with a good CPI

The idea is not landing: test a new theme for the same rules before more content.

Where the lines come from. The go thresholds are quoted from publishers' published guidance (sources below). For scale: across hyper-casual games in Q3 2022, the median Android D1 was 19% and the top 2% reached 42% [3]. Each publisher sets its own lines; these are a starting point, to be replaced by the publisher's own benchmarks.

[1] Supersonic: "The 3 most important KPIs for testing your hyper-casual prototype", supersonic.com/learn/blog/the-3-most-important-kpis-for-testing-your-hyper-casual-prototype

[2] Ivan Fedyanin (Ducky), PocketGamer.biz, 17 June 2021: "What are the key KPIs for a successful hypercasual game?", pocketgamer.biz/comment-and-opinion/76718/ducky-ivan-fedyanin-key-hypercasual-kpis

[3] Tenjin and GameAnalytics, Q3 2022: "Hypercasual benchmarks (via Game Dev Reports, 11 October 2022)", gamedevreports.substack.com/p/tenjin-and-gameanalytics-hypercasual

3 · KPI PLAN

What to measure, and what to change

The prototype has no analytics yet. These are the events to add, each tied to a question the design needs answered.

EVENT (TO ADD)	FIELDS	ANSWERS
`level_start`	level, attempt	Where players stop (the level funnel)
`level_win`	level, moves, par, stars, seconds, hints, restarts	Is the curve right? Moves over par per level
`dead_end`	level, moves	Are the bots' walls walls for people?
`move_refused`	level, reason (too tall · cart · locked · colour · full)	Which rule players haven't understood
`hint_used`	level, move_number	Which levels are walls, and at which move
`session_start`	day_since_install, level	Retention (D1, D7)

KPIs and targets

KPI	TARGET OR BENCHMARK	IF IT MISSES
D1 retention	≥ 38% Android [1]	The first minute: levels 1–5, the first tall volume (4)
D7 retention	Benchmark: the top 2% of hyper-casual reach 16% on Android [3]	Chapter pacing; the walls the funnel shows
Dead ends per level	Set from the first cohort	Too many: an extra empty shelf, or undo
"Too tall" refusals on levels 4–6	Set from the first cohort	Players don't read height: make tall spines and tall shelves more distinct
Moves over par	Compare with the bots (workbook)	Where people and bots disagree, people are right: retune

Watch these levels first

Where the greedy bot (workbook) dead-ends most often: **9 Reading Room (96%), 16 Mixed Heights (95%), 17 Full House (92%), 19 Grand Atlas (98%), 29 Overdue (100%), 36 Cart and Key (100%)**. Bots aren't people, but these are the first places to look in the dead-end and restart data.

Nothing guessed

Targets with a source carry its number; the rest are set from the first real cohort instead of guessed. The bots' results in the workbook are the baseline the real data is read against.

[1] Supersonic: "The 3 most important KPIs for testing your hyper-casual prototype", supersonic.com/learn/blog/the-3-most-important-kpis-for-testing-your-hyper-casual-prototype

[2] Ivan Fedyanin (Ducky), PocketGamer.biz, 17 June 2021: "What are the key KPIs for a successful hypercasual game?", pocketgamer.biz/comment-and-opinion/76718/ducky-ivan-fedyanin-key-hypercasual-kpis

[3] Tenjin and GameAnalytics, Q3 2022: "Hypercasual benchmarks (via Game Dev Reports, 11 October 2022)", gamedevreports.substack.com/p/tenjin-and-gameanalytics-hypercasual